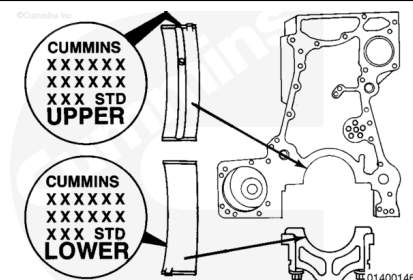


## Trainings ▾

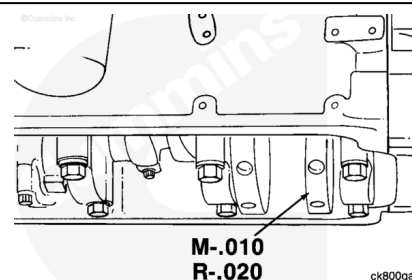
### General Information

**Note :** The upper bearings contain an oil hole. The lower bearings do not have an oil hole. Both bearings are marked on the back side indicating location (UPPER or LOWER) and size (STANDARD [STD] or OVERSIZE [OS]). The amount of OS is indicated in U.S. customary inches.

Use the same size bearing [STD, 0.010, 0.020, or 0.030] that was removed.

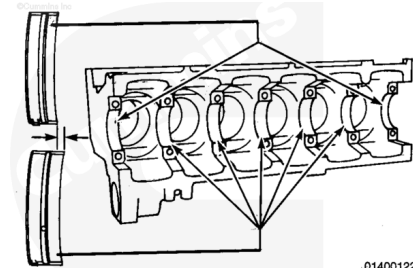


The crankshaft will be stamped on the end of the number one counterweight to indicate if it has been ground under size. Thrust bearing size is stamped on a crankshaft counterweight adjacent to the thrust location.



The location number of the main bearing caps, beginning at the front, are numbers 1 through 7.

The main bearings are two widths. The narrow main bearings fit locations number 1 and number 7. The wide bearings fit the remaining locations.



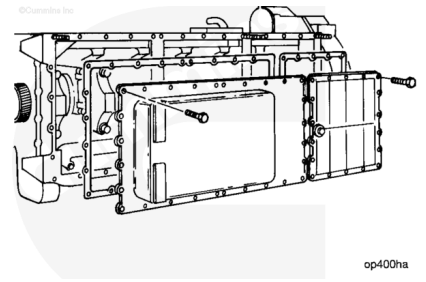
### Preparatory Steps

#### **⚠ WARNING ⚠**

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

### **⚠ WARNING ⚠**

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



### **⚠ WARNING ⚠**

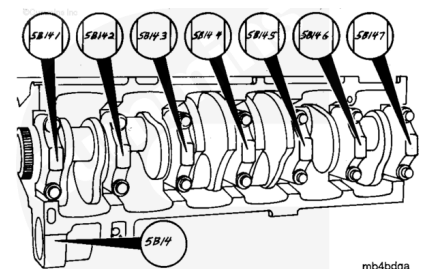
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/18/18-007-037-tr.html>).
- Remove the oil filter elements. Refer to Procedure 007-013 in Section 7 (</qs3/pubsys2/xml/en/procedures/18/18-007-013-tr.html>).
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/18/18-007-025-tr.html>).
- Remove the lubricating oil pan adapter cover. Refer to Procedure 007-026 in Section 7 (</qs3/pubsys2/xml/en/procedures/18/18-007-026-tr.html>).
- Remove the oil suction tube and oil pan adapter. Refer to Procedure 007-027 in Section 7. (</qs3/pubsys2/xml/en/procedures/18/18-007-027-tr.html>)

## Remove

**Note :** Before removing the main bearings, refer to General Information step in this procedure.

The main bearings caps **must** be marked for position. The last number on each main bearing cap identifies the location and position in the cylinder block.



The cylinder block and main bearing cap identification numbers **must** be identical.

Use a steel stamp and mark any main bearing cap that is **not** marked correctly.

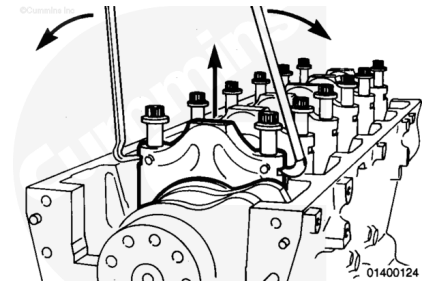
**⚠ WARNING ⚠**

**Remove the lower main bearings one pair at a time. Personal injury and damage to the crankshaft can result if the crankshaft falls.**

Loosen the main bearing cap capscrews until there is approximately 13 mm [ $\frac{1}{2}$  inch] between the capscrew head and the main bearing cap.

Install the pry bars in the notches at the sides of the main bearing cap and loosen it.

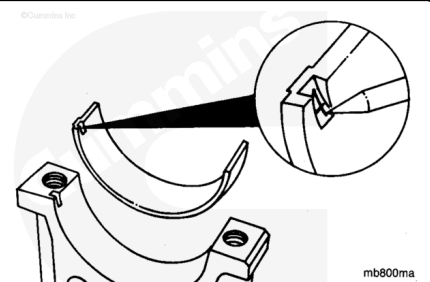
Remove the main bearing cap and capscrews.



Mark the bearing position for future identification or for possible failure analysis.

Remove the lower bearing. Use an awl and mark the bearing position in the tang area.

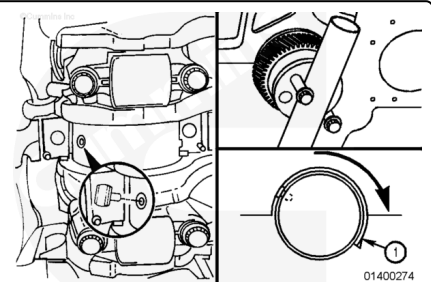
Remove the crankshaft thrust bearings from the number 6 main bearing cap and the number 6 saddle in the cylinder block. Refer to Procedure 001-007 in Section 1. ([/qs3/pubsys2/xml/en/procedures/18/18-001-007.html](http://qs3/pubsys2/xml/en/procedures/18/18-001-007.html))



To remove the upper main bearing shell install the main bearing roll out tool, Part Number 3823818, into the crankshaft main bearing journal oil hole.

Rotate the crankshaft so the tang (1) of the main bearing shell rolls out of the block first.

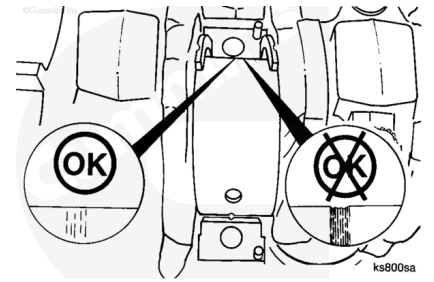
Use the crankshaft adapter mounting bolts and slowly turn the crankshaft until the bearing shell is out of the block.



## Clean and Inspect for Reuse

**⚠ WARNING ⚠**

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



**⚠ WARNING ⚠**

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

**⚠ CAUTION ⚠**

To reduce the possibility of bearing damage, do not use a scraper or a wire brush.

**Note :** Be sure the bearings are marked for location. The bearings must be installed in the original location if they are to be reused.

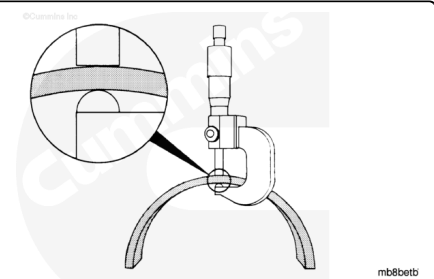
Use a lint-free cloth. Clean the crankshaft journals, and connecting rods, and main bearings.

Check the crankshaft journal for damage.

If the crankshaft journal is damaged, the crankshaft **must** be repaired or replaced.

Clean the bearings with solvent and dry with compressed air.

Use ball-end micrometer. Measure the bearing thickness at the wear location.



**Main Bearing Thickness Standard or Oversize (OS)**

|            | mm    |     | in     |
|------------|-------|-----|--------|
| Standard   | 4.280 | MIN | 0.1685 |
|            | 4.348 | MAX | 0.1712 |
| 0.010 (OS) | 4.407 | MIN | 0.1735 |
|            | 4.475 | MAX | 0.1762 |
| 0.020 (OS) | 4.534 | MIN | 0.1785 |
|            | 4.602 | MAX | 0.1812 |
| 0.030 (OS) | 4.661 | MIN | 0.1835 |
|            | 4.729 | MAX | 0.1862 |

|            |       |     |        |
|------------|-------|-----|--------|
| 0.040 (OS) | 4.788 | MIN | 0.1885 |
|            | 4.848 | MAX | 0.1907 |

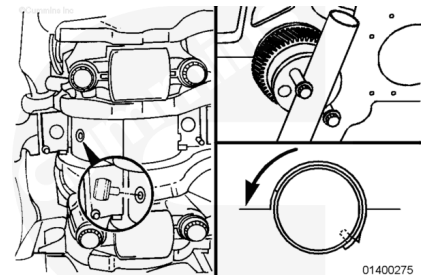
If the bearing is **not** within specifications, it **must** be replaced.

## Install

Used bearings **must** be installed in their original location.

Install the upper main bearing shells using the same method that was used for the removal of the shells.

The bearing tang (1) **must** fit into the slot (2) in the bearing saddle to assure proper location of the bearing.



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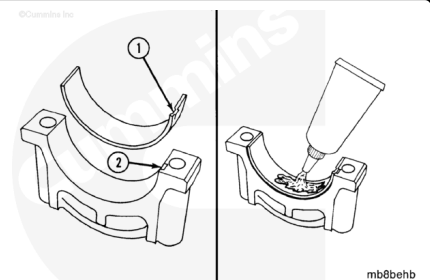
**Note :** Before installing the thrust bearings, refer to General Information step in this procedure and Procedure Refer to Procedure 001-007 in Section 1. (/qs3/pubsys2/xml/en/procedures/18/18-001-007.html)

Use a lint-free cloth. Clean the lower main bearings, the lower thrust bearings, and the mounting surfaces.

Do **not** lubricate the back of the main bearings.

Align the tang (1) in the bearing with the slot (2) in the main bearing cap. Install the bearing. The end of the bearing **must** be even with the main bearing cap mounting surface.

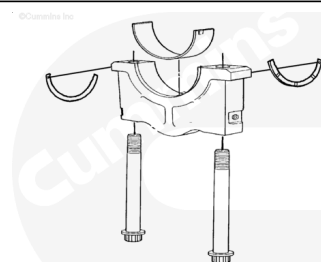
Lubricate the bearing surface with clean engine oil.



mb8behb

### **⚠ CAUTION ⚠**

**To reduce the possibility of engine damage, the grooves in the thrust bearings must point toward the crankshaft. The dowels that secure the bearings must not protrude above the bearing.**



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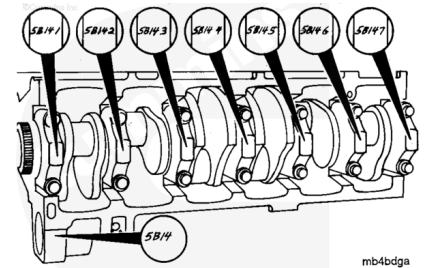
Lubricate the capscrew head and threads with SAE EP 140W oil. Allow the excess oil to drip off of the capscrews before installing in the block.

Install the capscrews in the cap.

Install the two thrust bearings on the number 6 main bearing cap. Refer to Procedure 001-007 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/18/18-001-007.html)

**⚠ CAUTION ⚠**

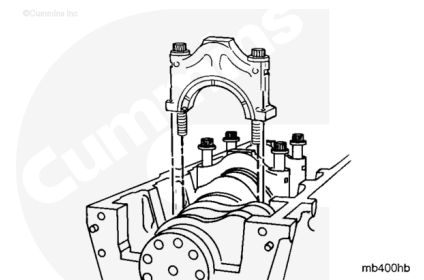
**To reduce the possibility of engine damage, the numbers on the main bearing caps must be the same as the numbers on the block.**



Check the number on the main bearing caps. The last digit of each number (1 through 7), indicates the correct location.

**⚠ CAUTION ⚠**

**To reduce the possibility of engine damage, make sure the side of the cap and bearing with the bearing locating tang is toward the tang in the block.**

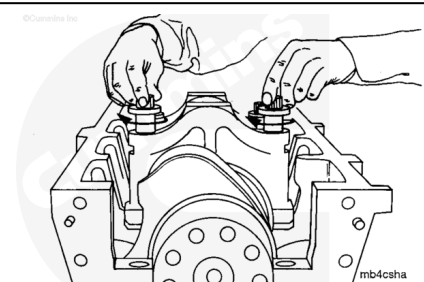


Lubricate the bearing surface with engine oil. The bearing shells **must** be firmly seated in the cap and the correct capscrew, washer combination in position. Install the main bearing caps.

**Note : Do not hit the main bearing caps with a hammer. The bearing shells can fall out.**

**⚠ CAUTION ⚠**

**Do not rotate the crankshaft until all of the main bearing caps are pulled to the block. Damage will result if the bearings move out-of-location.**



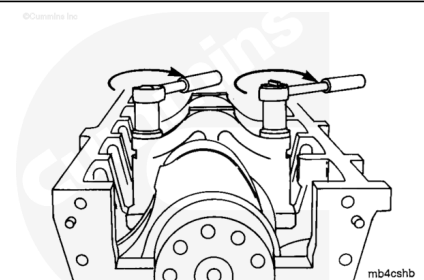
Install the capscrews.

Turn each capscrew until it touches the main bearing caps.

**Note : Do not use an impact wrench. The main bearing shells can fall out.**

Use both of the capscrews to pull the main bearing cap into position.

Use two wrenches and tighten both capscrews at the same time.

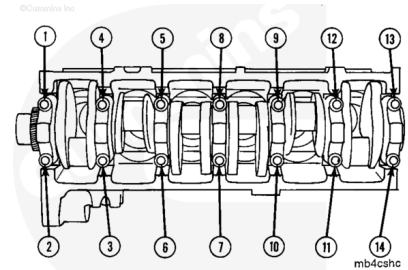


Check to be sure the number 7 cap is even with the back of the block.

Use the following steps and tighten the capscrews in the sequence shown.

**Torque Value:**

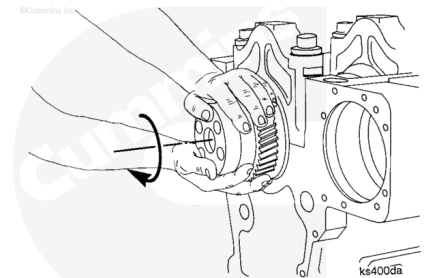
1. 265 n•m [ 195 ft-lb ]
2. 605 n•m [ 445 ft-lb ]
3. Loosen
4. 265 n•m [ 195 ft-lb ]
5. 605 n•m [ 445 ft-lb ]



If the pistons are removed, turn the crankshaft by hand.

If the pistons are installed, use the barring mechanism.

The crankshaft will turn freely if the main bearings are installed correctly.



## Finishing Steps

### **⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**

- Check the crankshaft end clearance. Refer to Procedure 001-016 in Section 1 (</qs3/pubsys2/xml/en/procedures/18/18-001-016-tr.html>).
- Install the lubricating oil pan adapter and the oil suction tube. Refer to Procedure 007-027 in Section 7 (</qs3/pubsys2/xml/en/procedures/18/18-007-027-tr.html>).
- Install the lubricating oil pan adapter cover. Refer to Procedure 007-026 in Section 7 (</qs3/pubsys2/xml/en/procedures/18/18-007-026-tr.html>).
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/18/18-007-025-tr.html>).
- Install new lubricating oil filter elements. Refer to Procedure 007-013 (</qs3/pubsys2/xml/en/procedures/18/18-007-013-tr.html>).
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/18/18-007-037-tr.html>).
- Operate the engine to 70°C [160°F] coolant temperature and check for leaks.

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**Last Modified: 04-Apr-2017**

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